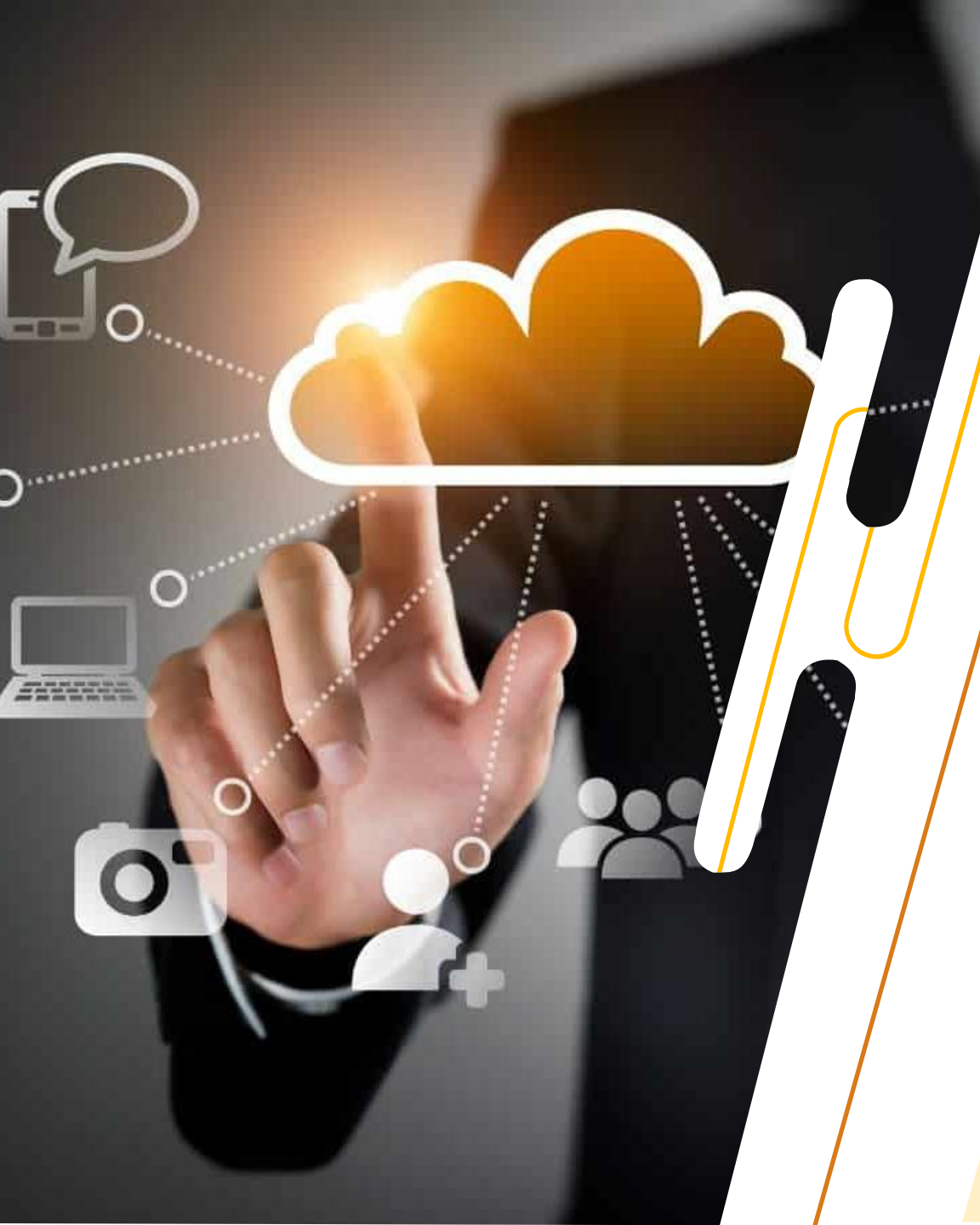




IT3061

Massive Data Processing and Cloud Computing

YEAR 3 SEMESTER 2
DATA SCIENCE SPECIALIZATION

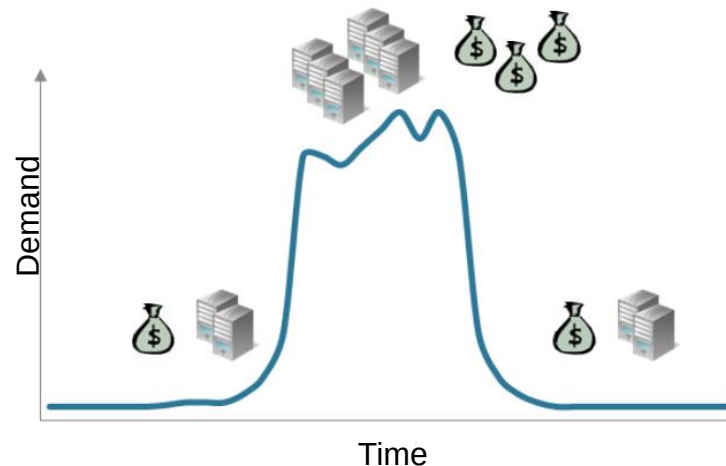
Lecture Outline

- 1. What is Elastic Cloud?
- 2. Elasticity Vs Scalability
- 3. Benefits of Elastic Cloud Computing
- 4. Cloud Compute Services



Elastic Computing

- Elastic computing (or cloud elasticity) is the ability to quickly expand or decrease computer processing, memory, and storage resources to meet changing demands without worrying about capacity planning and engineering for peak usage.
- System monitoring tools control Elastic computing so that the requirement of resources and the allocated resources go hand in hand and the operations are not disrupted.



Elasticity Vs Scalability

Cloud Elasticity	Cloud Scalability
Use to dynamically grow or shrink	Use to meet the static increase in the workload
Commonly used by small companies whose workload and demand increases only for a specific period of time	Used by large companies whose customer base persistently grows in order to do the operations efficiently
A short term planning and adopted just to deal with an unexpected increase in demand or seasonal demands	A long term planning and adopted just to deal with an expected increase in demand

Benefits of Elastic Cloud Computing

- Provide resizable compute capacity
 - Unlimited virtual servers as your own data center
 - Easy setup, start or stop VMs within minutes
 - Vertical and Horizontal Scaling
- Eliminate investment in computing hardware upfront
 - Pay-as-you go service
 - No capital expenditure
 - Cost efficient

Benefits of Elastic Cloud Computing

- Ability to add capacity automatically
 - Automatically scale up to add capacity to handle extra traffic
 - Automatically scale down to reduce capacity to handle reduced traffic

Elasticity Resources on AWS

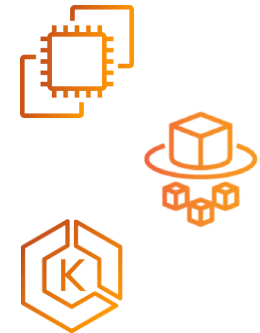
Some services do this as part of their service,

- Amazon S3
- Amazon SQS
- Amazon SNS
- Amazon SES
- Amazon Aurora



Others integrate with AWS Auto Scaling,

- Amazon EC2
- Amazon ECS
- AWS Fargate
- Amazon EKS
- Amazon DynamoDB



Some require vertical scaling,

- Amazon RDS



Cloud Compute Services

- AWS - Elastic Compute Cloud (EC2)
- Azure – Virtual Machines

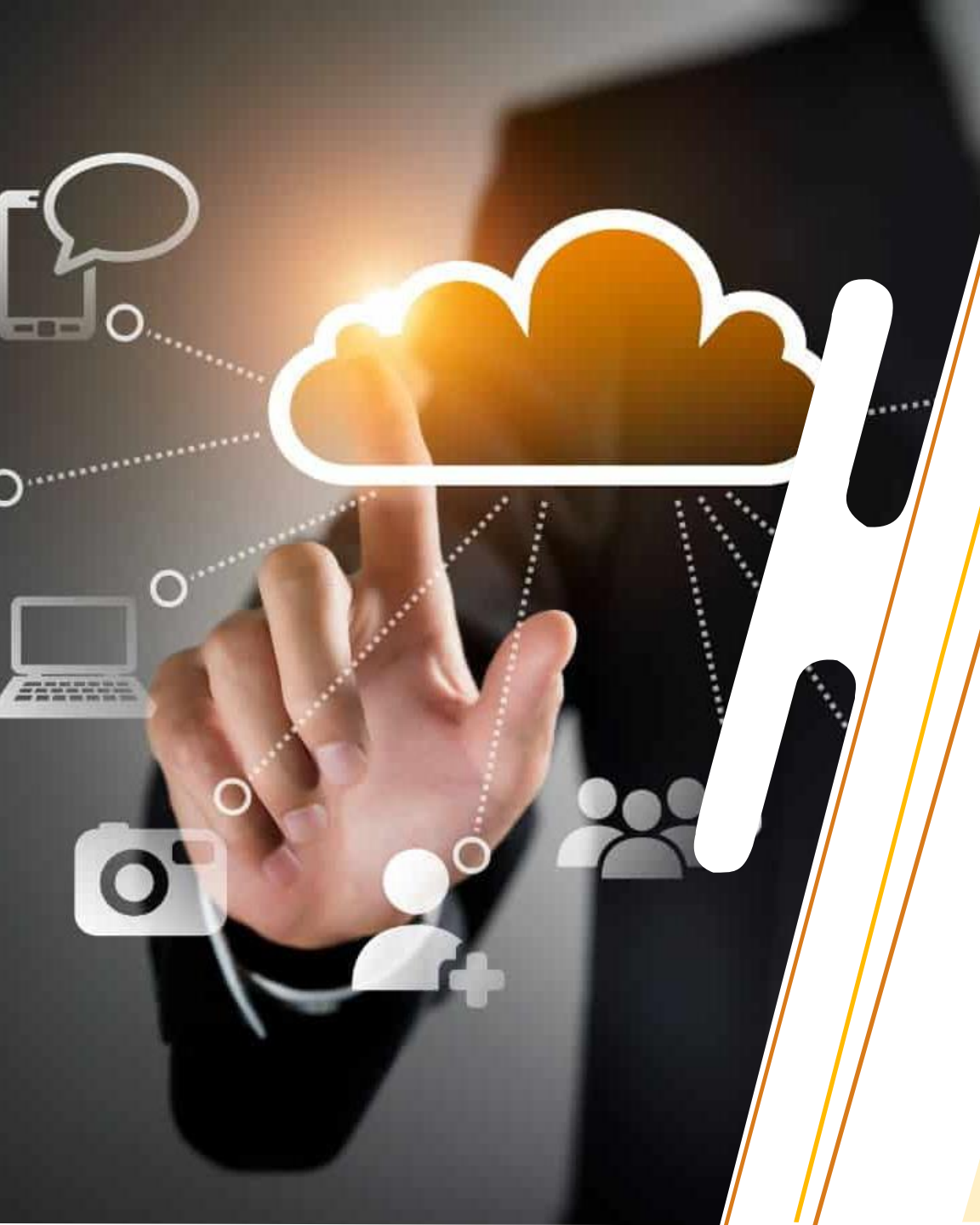


Amazon EC2





Demo



Next Session :
Cloud Services